

Annex 176B

(normative)

PMA instantiation options for 200 Gb/s, 400 Gb/s, 800 Gb/s, 1.6 Tb/s Physical Layer implementations

176B.1 Scope

This annex provides guidelines for implementing one or more 200GAUI-n, 400GAUI-n, 800GAUI-n, and 1.6TAUI-n within a PHY or xMII Extender.

176B.2 Introduction

Each 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R PHY is defined within the associated PMD clause to include PCS, PMA, and PMD sublayers. Some PHYs also include an Inner FEC sublayer in place of the PMA above the PMD (e.g., 800GBASE-FR4 defined in Clause 183 and 800GBASE-LR1 defined in Clause 185). Some other PHYs include an FEC (rather than the RS-FEC provided by the PCS) and PMA specific to the PMD (e.g., 800GBASE-ER1 defined in Clause 187). A set of generic examples are shown in Figure 176B–1. The PMA or Inner FEC below the PCS converts the number of PCS lanes (q) to the required number of PMD lanes (p).

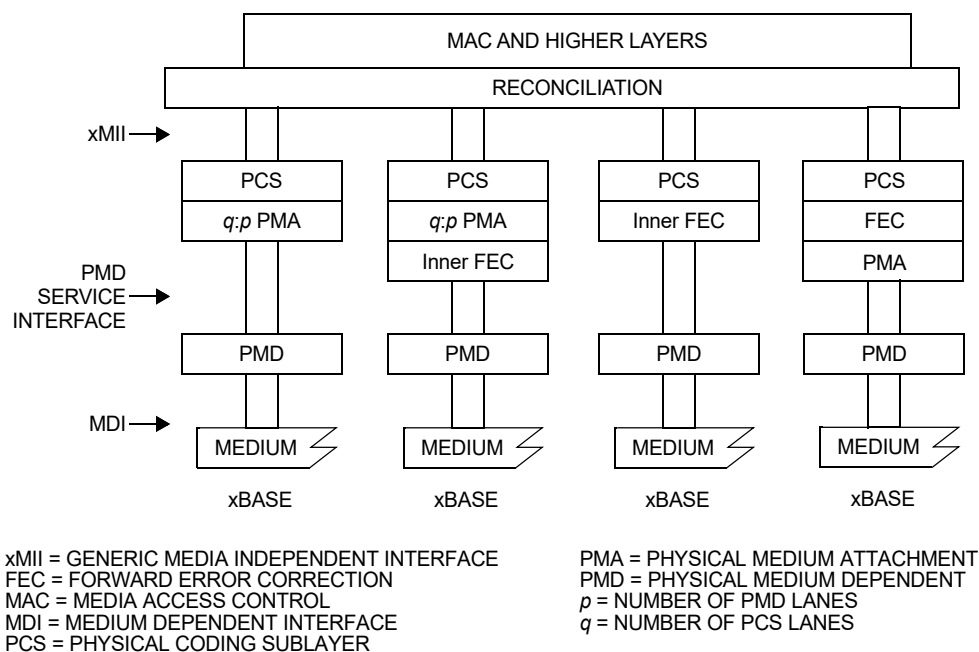


Figure 176B–1—PHYs without physically instantiated interfaces between the PCS and PMD

Within a 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R PHY, the interface between the PCS and the PMD may be extended by means of one or two physically instantiated interfaces of

200GAUI-n, 400GAUI-n, 800GAUI-n, and 1.6TAUI-n, respectively, where “n” indicates the number of physical lanes. Each xAUI-n is bookended by a pair of PMAs which convert the number of lanes between the sublayer or interface above to the sublayer or interface below.

A set of generic examples of PHYs with one or two xAUI-n are shown in Figure 176B–2. The $q:n$ or $q:m$ PMA below the PCS converts between the number of PCS lanes (q) and the number of xAUI-n lanes (m or n). The $n:p$ PMA above the PMD converts between the number of xAUI-n lanes and the number of PMD lanes (p). The $m:n$ PMA between two xAUI-n converts between the number of upper xAUI-n lanes (m) and number of lower xAUI-n lanes (n). This set of PMA and xAUI-n instances, is functionally equivalent to the $q:p$ PMA used in xBASE-R PHYs without xAUI-n as shown in Figure 176B–1.

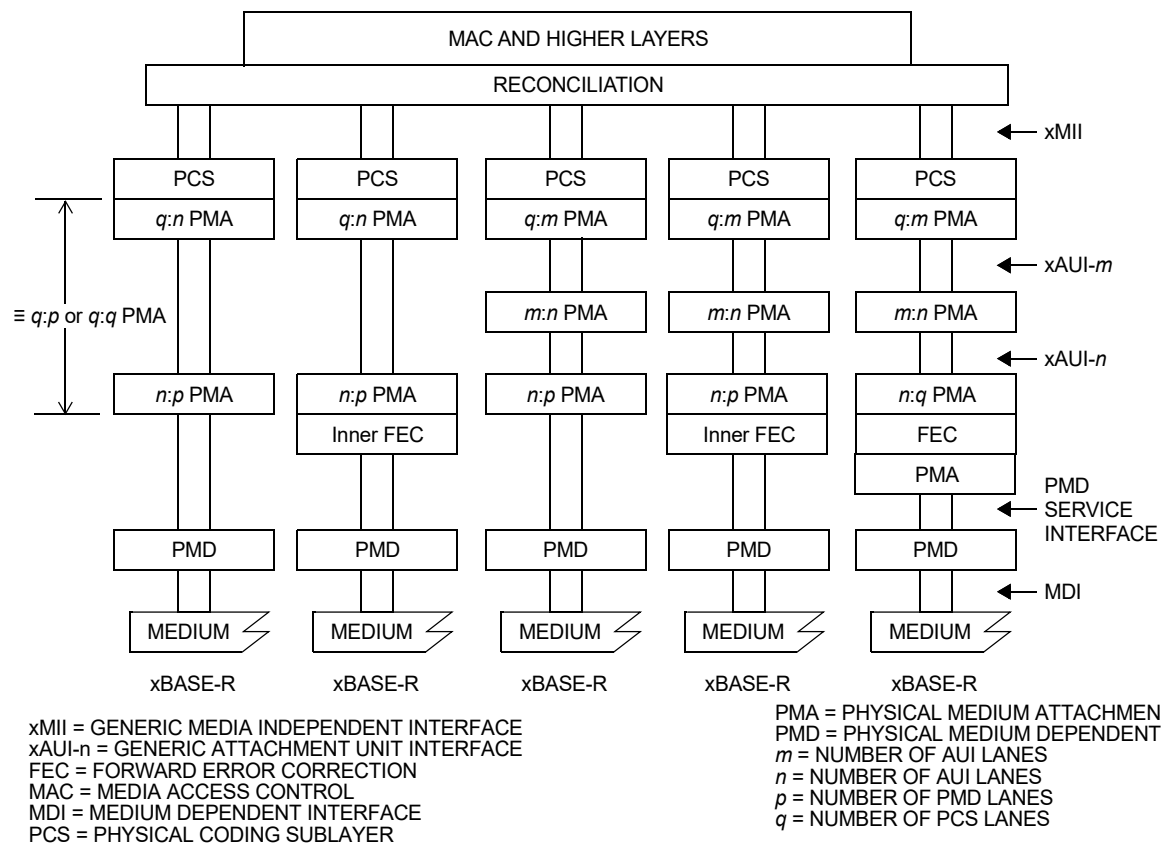


Figure 176B–2—xBASE-R PHYs with one and two physically instantiated interfaces

Within a 200 Gb/s, 400 Gb/s, 800 Gb/s, or 1.6 Tb/s Physical Layer implementation, the interface between the MAC/RS and the PCS may be extended by means of a 200GMII, 400GMII, 800GMII, or 1.6TMII Extender. An xMII Extender includes one or two physically instantiated interfaces 200GAUI-n, 400GAUI-n, 800GAUI-n, and 1.6TAUI-n, where “n” indicates the number of physical lanes. Each xAUI-n is bookended by a pair of PMAs, which convert the number of lanes between the sublayer or interface above to the sublayer or interface below.

A set of generic examples of Physical Layer implementations including a xMII Extender with one or two xAUI-n are shown in Figure 176B–3. The $q:n$ or $q:m$ PMA below the DTE XS converts between the number of PCS lanes (q) and the number of xAUI-n lanes (m or n). The $n:q$ PMA above the PHY XS converts

between the number of xAUI- n lanes and the number of PCS lanes (q). The $m:n$ PMA between two xAUI- n converts between the number of upper xAUI- n lanes (m) and number of lower xAUI- n lanes (n). This set of PMA and xAUI- n instances, is functionally equivalent to a $q:q$ PMA.

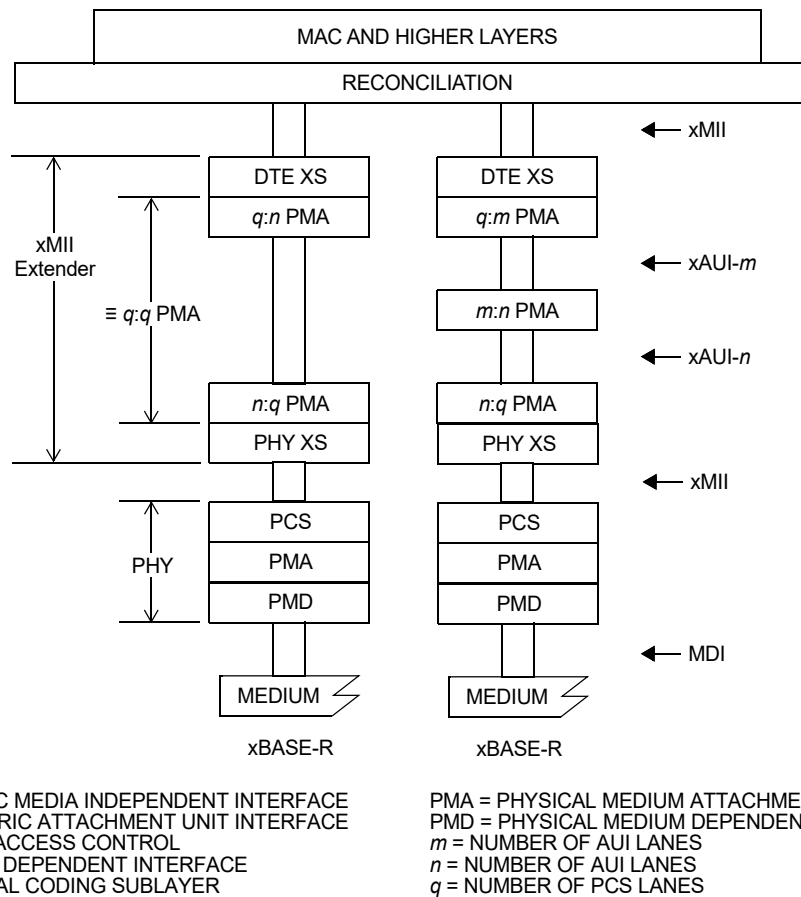


Figure 176B-3—xBASE Physical Layer implementations with an xMII Extender with one and two physically instantiated interfaces

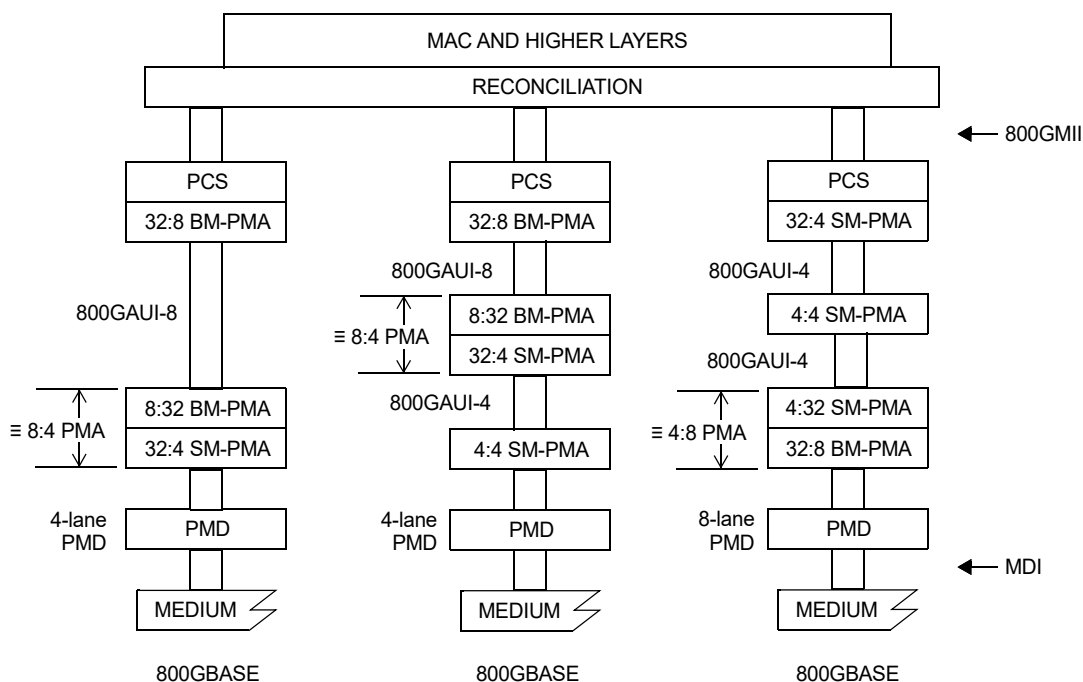
Generally, a set of xAUI- n and PMA within an xMII Extender or within a xBASE-R PHY may be referred to as a PMA instantiation. Particular forms of a PMA instantiation may be denoted as follows. A $q:n:p$ (or $q:n:q$) PMA instantiation has q PCS lanes at the top, one xAUI- n with n lanes, and either q PCS lanes or p PMD lanes below. Similarly, a $q:m:n:p$ (or $q:m:n:q$) PMA instantiation has q PCS lanes at the top, one xAUI- n with m lanes, a second xAUI- n with n lanes, and either q PCS lanes or p PMD lanes below.

The $q:n:p$ and $q:m:n:p$ PMA instantiations may take various forms depending upon the number of PCS lanes, the number of lanes on each xAUI- n , and the number of PMD lanes. Details of each supported form are described in 176B.5 through 176B.8.

176B.3 Special Case for 200GBASE-R, 400GBASE-R, and 800GBASE-R PMAs

For 200GBASE-R, 400GBASE-R, and 800GBASE-R PMA, two types of lane multiplexing are defined. Bit multiplexing PMA (BM-PMA) types are defined in Clause 120 and Clause 173. Symbol multiplexing PMA

(SM-PMA) types are defined in Clause 176. A BM-PMA is used to attach to an xAUI-n or PMD with lane data rates of 100 Gb/s or less while an SM-PMA is used to attach to an xAUI-n or PMD with lane data rates of 200 Gb/s. In order to convert between symbol multiplexing and bit multiplexing a BM-PMA and SM-PMA are placed back to back with each PMA demultiplexing to PCS lanes between. Some examples for 800GBASE-R PHYs are shown in Figure 176B-4. In these examples, an 8:32 BM-PMA combined with a 32:4 SM-PMA is equivalent in function to an 8:4 PMA. In other words, an 8:4 PMA may be formed from the combination of an 8:32 BM-PMA and a 32:4 SM-PMA.



NOTE---400GBASE-R and 200GBASE-R PHYs are formed in a similar way except that the number of lanes is divided by two and four, respectively.

800GMII = 800 Gb/s MEDIA INDEPENDENT INTERFACE
800GAUI = 800 Gb/s ATTACHMENT UNIT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

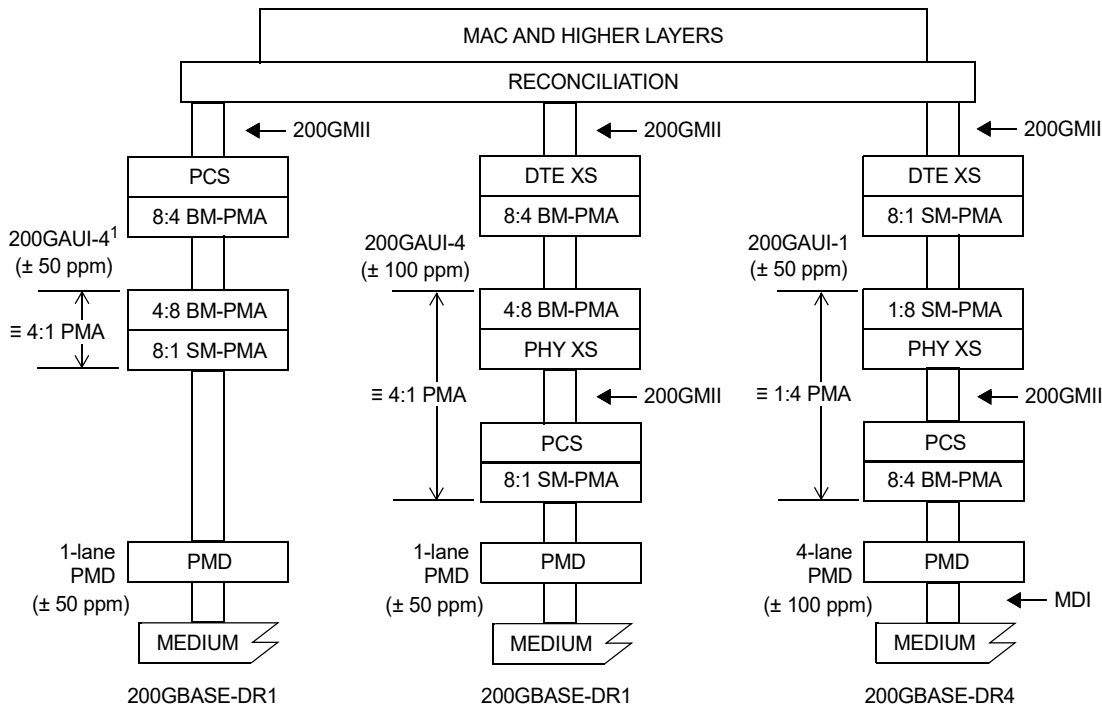
PCS = PHYSICAL CODING SUBLAYER
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 176B-4—800GBASE-R PHYs with back to back PMA

176B.4 Special case for 200BASE-R and 400GBASE-R Physical Layers with 100 ppm signaling

For 200GBASE-R and 400GBASE-R Physical Layers, the signaling rate range for bit-multiplexed lanes in some cases is specified as ± 100 ppm while the signaling rate range for symbol-multiplexed lanes is specified as ± 50 ppm. In these cases an xMII Extender (see Clause 118) with rate matching is required to convert from a signaling rate range of ± 100 ppm to ± 50 ppm.

Examples of 200GBASE-R and 400GBASE-R Physical Layers with a mix of ± 100 ppm and ± 50 ppm signaling domains are shown in Figure 176B-5 and Figure 176B-6.



¹ 200GAUI-4 where the signaling rate range is limited to ± 50 ppm instead of ± 100 ppm.

200GMII = 200 Gb/s MEDIA INDEPENDENT INTERFACE
200GAUI = 200 Gb/s ATTACHMENT UNIT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 176B-5— Examples of 200GBASE-R Physical Layer implementations including the use of a 200GMII Extender for signaling rate range matching

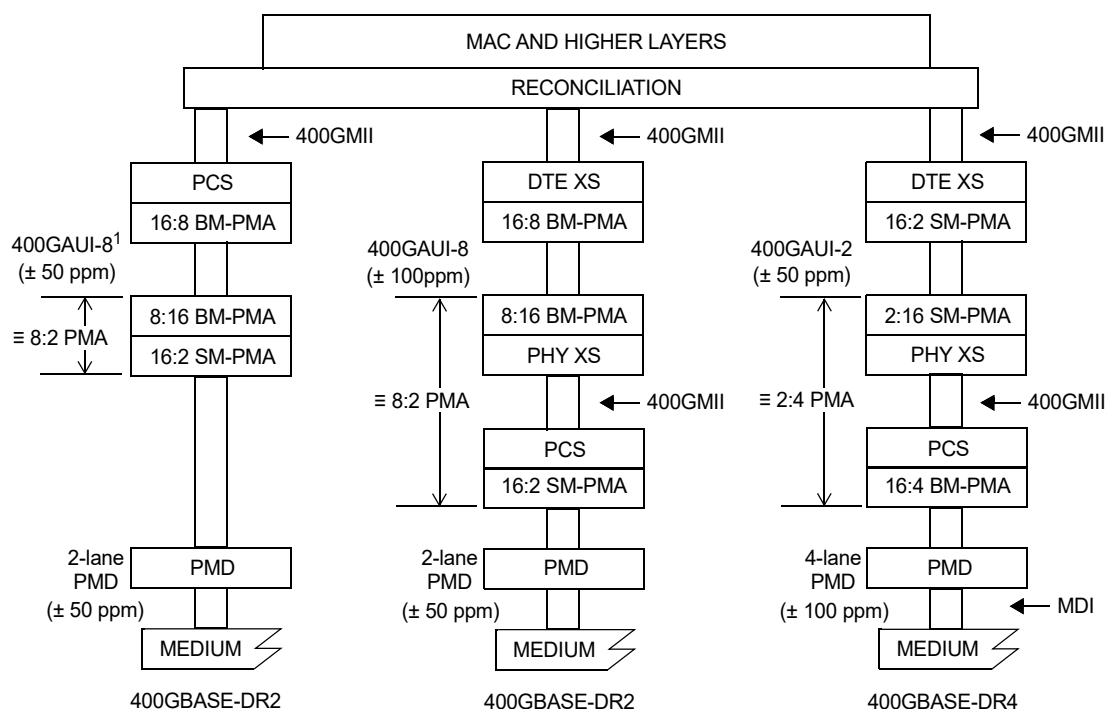


Figure 176B-6—Examples of 400GBASE-R Physical Layer implementations showing the use of a 400GMII Extender for signal rate range matching

176B.5 PMA instantiation for 200 Gb/s Physical Layer Implementations

176B.5.1 8:4, 8:2 and 8:1 PMA instantiations for 200GBASE-R PHYs

PMA instantiations for use within a 200GBASE-R PHY are shown in Figure 176B-2, where the number of PCS lanes q is 8. Instantiations may be of the form $8:n:p$ and $8:m:n:p$, where p is the number PMD lanes. Specific implementations of the PMA instantiations are defined in Table 176B-1 through Table 176B-6. These instantiations are relevant to the 200GBASE-R PHY types listed in Table 116-3, Table 116-4, and Table 116-4a, where the number of PMD lanes p is 1, 2, or 4.

These instantiations include C2M and C2C 200GAUI- n defined in Annex 120B, Annex 120C, Annex 120D, Annex 120E, Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUl lanes (m or n) is 1, 2, 4, or 8. Interfaces with 2, 4, or 8 lanes require a BM-PMA and interfaces with 1 lane require an SM-PMA. Each table specifies combinations of symbol-multiplexed interfaces (“S”) and bit-multiplexed interfaces (“B”) noted in each figure title. For example, an instantiation with a two S 200GAUI- n and a B PMD is denoted “SSB”.

Table 176B–1—200 Gb/s PMA 8:{2,4,8}:{2,4} and 8:1:1 (SS or BB) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>n</i> PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120B (C2C), 120C (C2M) <i>n</i> = 4: 120D (C2C), 120E (C2M) <i>n</i> = 2: 120F (C2C), 120G (C2M) <i>n</i> = 1: 176C (C2C), 176D (C2M)	
200GBASE-R <i>n</i> : <i>p</i> PMA	<i>p</i> = 2 or 4: 120 (BM-PMA) <i>p</i> = 1: 176 (SM-PMA)	

Table 176B–2—200 Gb/s PMA 8:1:{2,4} and 8:{2,4,8}:1 (SB or BS) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>n</i> PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120B (C2C), 120C (C2M) <i>n</i> = 4: 120D (C2C), 120E (C2M) <i>n</i> = 2: 120F (C2C), 120G (C2M) <i>n</i> = 1: 176C (C2C), 176D (C2M)	
200GBASE-R <i>n</i> :8 PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	The combination of <i>n</i> :8 PMA and 8: <i>p</i> PMA forms an <i>n</i> : <i>p</i> PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: <i>p</i> PMA	<i>p</i> = 2 or 4: 120 (BM-PMA) <i>p</i> = 1: 176 (SM-PMA)	

Table 176B–3—200 Gb/s PMA 8:{2,4,8}:{2,4,8}:{2,4} and 8:1:1:1 (BBB or SSS) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>m</i> PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	
200GAUI- <i>m</i>	<i>m</i> = 8: 120B (C2C) <i>m</i> = 4: 120D (C2C) <i>m</i> = 2: 120F (C2C) <i>m</i> = 1: 176C (C2C)	
200GBASE-R <i>m</i> : <i>n</i> PMA	{ <i>m</i> , <i>n</i> } = 2, 4, or 8: 120 (BM-PMA) { <i>m</i> , <i>n</i> } = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120C (C2M) <i>n</i> = 4: 120E (C2M) <i>n</i> = 2: 120G (C2M) <i>n</i> = 1: 176D (C2M)	
200GBASE-R <i>n</i> : <i>p</i> PMA	<i>p</i> = 2 or 4: 120 (BM-PMA) <i>p</i> = 1: 176 (SM-PMA)	

Table 176B–4—200 Gb/s PMA 8:1:1:{2,4}, 8:{2,4,8}:{2,4,8}:1 (BBS or SSB) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>m</i> PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	
200GAUI- <i>m</i>	<i>m</i> = 8: 120B (C2C) <i>m</i> = 4: 120D (C2C) <i>m</i> = 2: 120F (C2C) <i>m</i> = 1: 176C (C2C)	
200GBASE-R <i>m</i> : <i>n</i> PMA	{ <i>m</i> , <i>n</i> } = 2, 4, or 8: 120 (BM-PMA) { <i>m</i> , <i>n</i> } = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120C (C2M) <i>n</i> = 4: 120E (C2M) <i>n</i> = 2: 120G (C2M) <i>n</i> = 1: 176D (C2M)	
200GBASE-R <i>n</i> :8 PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	The combination of <i>n</i> :8 PMA and 8: <i>p</i> PMA forms an <i>n</i> : <i>p</i> PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: <i>p</i> PMA	<i>p</i> = 2 or 4: 120 (BM-PMA) <i>p</i> = 1: 176 (SM-PMA)	

Table 176B–5—200 Gb/s PMA 8:1:{2,4,8}:{2,4}, 8:{2,4,8}:1:1 (BSS or SBB) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>m</i> PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	
200GAUI- <i>m</i>	<i>m</i> = 8: 120B (C2C) <i>m</i> = 4: 120D (C2C) <i>m</i> = 2: 120F (C2C) <i>m</i> = 1: 176C (C2C)	
200GBASE-R <i>m</i> :8 PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	The combination of <i>m</i> :8 PMA and 8: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: <i>n</i> PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120C (C2M) <i>n</i> = 4: 120E (C2M) <i>n</i> = 2: 120G (C2M) <i>n</i> = 1: 176D (C2M)	
200GBASE-R <i>n</i> : <i>p</i> PMA	<i>p</i> = 2 or 4: 120 (BM-PMA) <i>p</i> = 1: 176 (SM-PMA)	

Table 176B–6—200 Gb/s PMA 8:{2,4,8}:1:{2,4} and 8:1:{2,4,8}:1 (SBS or BSB) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>m</i> PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	
200GAUI- <i>m</i>	<i>m</i> = 8: 120B (C2C) <i>m</i> = 4: 120D (C2C) <i>m</i> = 2: 120F (C2C) <i>m</i> = 1: 176C (C2C)	
200GBASE-R <i>m</i> :8 PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	The combination of <i>m</i> :8 PMA and 8: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: <i>n</i> PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120C (C2M) <i>n</i> = 4: 120E (C2M) <i>n</i> = 2: 120G (C2M) <i>n</i> = 1: 176D (C2M)	
200GBASE-R <i>n</i> :8 PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	The combination of <i>n</i> :8 PMA and 8: <i>p</i> PMA forms an <i>n</i> : <i>p</i> PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: <i>p</i> PMA	<i>p</i> = 2 or 4: 120 (BM-PMA) <i>p</i> = 1: 176 (SM-PMA)	

176B.5.2 8:8 PMA instantiations for 200GMII Extender

PMA instantiations for use within an 200GMII Extender (see Clause 118) are shown in Figure 176B–3, where the number of PCS lanes q is 8. Instantiations may be of the form $8:n:8$ and $8:m:n:8$. Specific implementations of the PMA instantiations are defined in Table 176B–7 through Table 176B–9.

These instantiations include C2M and C2C 200GAUI- n defined in Annex 120B, Annex 120C, Annex 120D, Annex 120E, Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUI lanes (m or n) is 1, 2, 4, or 8. Interfaces with 2, 4, or 8 lanes require a BM-PMA and interfaces with 1 lane require an SM-PMA. Each table specifies combinations of symbol-multiplexed interfaces (“S”) and bit-multiplexed interfaces (“B”) noted in each figure title. For example, an instantiation with a one S 200GAUI- n and one B 200GAUI- n is denoted “SB” or “BS”.

Table 176B–7—200 Gb/s 8:{2,4,8}:8 and 8:1:8 (S or B) PMA instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: n PMA	$n = 2, 4, \text{ or } 8$: 120 (BM-PMA) $n = 1$: 176 (SM-PMA)	
200GAUI- n	$n = 8$: 120B (C2C), 120C (C2M) $n = 4$: 120D (C2C), 120E (C2M) $n = 2$: 120F (C2C), 120G (C2M) $n = 1$: 176C (C2C), 176D (C2M)	
200GBASE-R $n:8$ PMA	$n = 2, 4, \text{ or } 8$: 120 (BM-PMA) $n = 1$: 176 (SM-PMA)	

Table 176B–8—200 Gb/s 8:{2,4,8}:{2,4,8}:8 and 8:1:1:8 (SS or BB) PMA instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: m PMA	$m = 2, 4, \text{ or } 8$: 120 (BM-PMA) $m = 1$: 176 (SM-PMA)	
200GAUI- m	$m = 8$: 120B (C2C) $m = 4$: 120D (C2C) $m = 2$: 120F (C2C) $m = 1$: 176C (C2C)	
200GBASE-R $m:n$ PMA	$\{m, n\} = 2, 4, \text{ or } 8$: 120 (BM-PMA) $\{m, n\} = 1$: 176 (SM-PMA)	
200GAUI- n	$n = 8$: 120C (C2M) $n = 4$: 120E (C2M) $n = 2$: 120G (C2M) $n = 1$: 176D (C2M)	
200GBASE-R $n:8$ PMA	$n = 2, 4, \text{ or } 8$: 120 (BM-PMA) $n = 1$: 176 (SM-PMA)	

Table 176B–9—200 Gb/s PMA 8:1:{2,4,8}:8 and 8:{2,4,8}:1:8 (SB or BS) instantiations

Sublayer	Clause/Annex	Notes
200GBASE-R 8: <i>m</i> PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	
200GAUI- <i>m</i>	<i>m</i> = 8: 120B (C2C) <i>m</i> = 4: 120D (C2C) <i>m</i> = 2: 120F (C2C) <i>m</i> = 1: 176C (C2C)	
200GBASE-R <i>m</i> :8 PMA	<i>m</i> = 2, 4, or 8: 120 (BM-PMA) <i>m</i> = 1: 176 (SM-PMA)	The combination of <i>m</i> :8 PMA and 8: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing.
200GBASE-R 8: <i>n</i> PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	
200GAUI- <i>n</i>	<i>n</i> = 8: 120C (C2M) <i>n</i> = 4: 120E (C2M) <i>n</i> = 2: 120G (C2M) <i>n</i> = 1: 176D (C2M)	
200GBASE-R <i>n</i> :8 PMA	<i>n</i> = 2, 4, or 8: 120 (BM-PMA) <i>n</i> = 1: 176 (SM-PMA)	

176B.6 PMA instantiation for 400 Gb/s Physical Layer Implementations

176B.6.1 16:16, 16:8, 16:4, and 16:2 PMA instantiations for 400GBASE-R PHYs

PMA instantiations for use within a 400GBASE-R PHY are shown in Figure 176B–2, where the number of PCS lanes *q* is 16. Instantiations may be of the form 16:*n*:*p* and 16:*m*:*n*:*p*, where *p* is the number PMD lanes. Specific implementations of the PMA instantiations are defined in Table 176B–10 through Table 176B–15. These instantiations are relevant to the 400GBASE-R PHY types listed in Table 116–3a, Table 116–5, and Table 116–5a, where the number of lanes *p* is 2, 4, 8, or 16.

These instantiations include C2M and C2C 400GAUI-*n* defined in Annex 120B, Annex 120C, Annex 120D, Annex 120E, Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUl lanes (*m* or *n*) is 2, 4, 8, or 16. Interfaces with 4, 8, or 16 lanes require a BM-PMA and interfaces with 2 lanes require an SM-PMA. Each table specifies combinations of symbol-multiplexed interfaces (“S”) and bit-multiplexed interfaces (“B”) noted in each figure title. For example, an instantiation with a two S 400GAUI-*n* and one B PMD is denoted “SSB”.

Table 176B–10—400 Gb/s PMA 16:{4,8,16}:{4,8,16}, 16:4:4 and 16:2:2 (SS or BB) instantiation

Sublayer	400 Gb/s Clause/Annex	Notes
400GBASE-R 16: <i>n</i> PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120B (C2C), 120C (C2M) <i>n</i> = 8: 120D (C2C), 120E (C2M) <i>n</i> = 4: 120F (C2C), 120G (C2M) <i>n</i> = 2: 176C (C2C), 176D (C2M)	
400GBASE-R <i>n</i> : <i>p</i> PMA	<i>p</i> = 4, 8, or 16: 120 (BM-PMA) <i>p</i> = 2: 176 (SM-PMA)	

Table 176B–11—400 Gb/s PMA 16:2:{4,8,16} and 16:{4,8,16}:2 (BS or SB) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>n</i> PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120B (C2C), 120C (C2M) <i>n</i> = 8: 120D (C2C), 120E (C2M) <i>n</i> = 4: 120F (C2C), 120G (C2M) <i>n</i> = 2: 176C (C2C), 176D (C2M)	
400GBASE-R <i>n</i> :16 PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	The combination of <i>n</i> :16 PMA and 16: <i>p</i> PMA forms an <i>n</i> : <i>p</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>p</i> PMA	<i>p</i> = 4, 8, or 16: 120 (BM-PMA) <i>p</i> = 2: 176 (SM-PMA)	

Table 176B–12—400 Gb/s PMA 16:{4,8,16}:{4,8,16} and 16:2:2:2 (BBB or SSS) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>m</i> PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	
400GAUI- <i>m</i>	<i>m</i> = 16: 120B (C2C) <i>m</i> = 8: 120D (C2C) <i>m</i> = 4: 120F (C2C) <i>m</i> = 2: 176C (C2C)	
400GBASE-R <i>m</i> : <i>n</i> PMA	{ <i>m</i> , <i>n</i> } = 4, 8, or 16: 120 (BM-PMA) { <i>m</i> , <i>n</i> } = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120C (C2M) <i>n</i> = 8: 120E (C2M) <i>n</i> = 4: 120G (C2M) <i>n</i> = 2: 176D (C2M)	
400GBASE-R <i>n</i> : <i>p</i> PMA	{ <i>n</i> , <i>p</i> } = 4, 8, or 16: 120 (BM-PMA) { <i>n</i> , <i>p</i> } = 2: 176 (SM-PMA)	

Table 176B–13—400 Gb/s PMA 16:2:2:{4,8,16} and 16:{4,8,16}:{4,8,16}:2 (BBS or SSB) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>m</i> PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	
400GAUI- <i>m</i>	<i>m</i> = 16: 120B (C2C) <i>m</i> = 8: 120D (C2C) <i>m</i> = 4: 120F (C2C) <i>m</i> = 2: 176C (C2C)	
400GBASE-R <i>m</i> : <i>n</i> PMA	{ <i>m</i> , <i>n</i> } = 4, 8, or 16: 120 (BM-PMA) { <i>m</i> , <i>n</i> } = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120C (C2M) <i>n</i> = 8: 120E (C2M) <i>n</i> = 4: 120G (C2M) <i>n</i> = 2: 176D (C2M)	
400GBASE-R <i>n</i> :16 PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	The combination of <i>n</i> :16 PMA and 16: <i>p</i> PMA forms an <i>n</i> : <i>p</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>p</i> PMA	<i>p</i> = 4, 8, or 16: 120 (BM-PMA) <i>p</i> = 2: 176 (SM-PMA)	

Table 176B–14—400 Gb/s PMA 16:2:{4,8,16}:{4,8,16} and 16:{4,8,16}:2:2 (SBB or BSS) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>m</i> PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	
400GAUI- <i>m</i>	<i>m</i> = 16: 120B (C2C) <i>m</i> = 8: 120D (C2C) <i>m</i> = 4: 120F (C2C) <i>m</i> = 2: 176C (C2C)	
400GBASE-R <i>m</i> :16 PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	The combination of <i>m</i> :16 PMA and 16: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>n</i> PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120C (C2M) <i>n</i> = 8: 120E (C2M) <i>n</i> = 4: 120G (C2M) <i>n</i> = 2: 176D (C2M)	
400GBASE-R <i>n</i> : <i>p</i> PMA	{ <i>n</i> , <i>p</i> } = 4, 8, or 16: 120 (BM-PMA) { <i>n</i> , <i>p</i> } = 2: 176 (SM-PMA)	

Table 176B–15—400 Gb/s PMA and 16:{4,8,16}:2:{4,8,16} and 16:2:{4,8,16}:2 (BSB or SBS) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>m</i> PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	
400GAUI- <i>m</i>	<i>m</i> = 16: 120B (C2C) <i>m</i> = 8: 120D (C2C) <i>m</i> = 4: 120F (C2C) <i>m</i> = 2: 176C (C2C)	
400GBASE-R <i>m</i> :16 PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	The combination of <i>m</i> :16 PMA and 16: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>n</i> PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120C (C2M) <i>n</i> = 8: 120E (C2M) <i>n</i> = 4: 120G (C2M) <i>n</i> = 2: 176D (C2M)	
400GBASE-R <i>n</i> :16 PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	The combination of <i>n</i> :16 PMA and 16: <i>p</i> PMA forms an <i>n</i> : <i>p</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>p</i> PMA	<i>p</i> = 4, 8, or 16: 120 (BM-PMA) <i>p</i> = 2: 176 (SM-PMA)	

176B.6.2 16:16 PMA instantiations for 400GMII Extender

PMA instantiations for use within an 400GMII Extender (see Clause 118) are shown in Figure 176B–3, where the number of PCS lanes *q* is 16. Instantiations may be of the form 16:*n*:16 and 16:*m*:*n*:16. Specific implementations of the PMA instantiations are defined in Table 176B–16 through Table 176B–18.

These instantiations include C2M and C2C 400GAUI-*n* defined in Annex 120B, Annex 120C, Annex 120D, Annex 120E, Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUl lanes (*m* or *n*) is 2, 4, 8, or 16. Interfaces with 4, 8, or 16 lanes require a BM-PMA and interfaces with 2 lanes require an SM-PMA. Each table specifies combinations of symbol-multiplexed interfaces (“S”) and bit-multiplexed interfaces (“B”) noted in each figure title. For example, an instantiation with a one S 400GAUI-*n* and one B 400GAUI-*n* is denoted “SB” or “BS”.

Table 176B–16—400 Gb/s and 16:{4,8,16}:16 and 16:2:16 (B or S) PMA instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>n</i> PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120B (C2C), 120C (C2M) <i>n</i> = 8: 120D (C2C), 120E (C2M) <i>n</i> = 4: 120F (C2C), 120G (C2M) <i>n</i> = 2: 176C (C2C), 176D (C2M)	
400GBASE-R <i>n</i> :16 PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	

Table 176B–17—400 Gb/s 16:{4,8,16}:{4,8,16}:16 and 16:2:2:16 (SS or BB) PMA instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>m</i> PMA	<i>m</i> = 4, 8 or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	
400GAUI- <i>m</i>	<i>m</i> = 16: 120B (C2C) <i>m</i> = 8: 120D (C2C) <i>m</i> = 4: 120F (C2C) <i>m</i> = 2: 176C (C2C)	
400GBASE-R <i>m</i> : <i>n</i> PMA	<i>{m, n}</i> = 4, 8, or 16: 120 (BM-PMA) <i>{m, n}</i> = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120C (C2M) <i>n</i> = 8: 120E (C2M) <i>n</i> = 4: 120G (C2M) <i>n</i> = 2: 176D (C2M)	
400GBASE-R <i>n</i> :16 PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	

Table 176B–18—400 Gb/s PMA 16:2:{4,8,16}:16 and 16:{4,8,16}:2:16 (SB or BS) instantiations

Sublayer	Clause/Annex	Notes
400GBASE-R 16: <i>m</i> PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	
400GAUI- <i>m</i>	<i>m</i> = 16: 120B (C2C) <i>m</i> = 8: 120D (C2C) <i>m</i> = 4: 120F (C2C) <i>m</i> = 2: 176C (C2C)	
400GBASE-R <i>m</i> :16 PMA	<i>m</i> = 4, 8, or 16: 120 (BM-PMA) <i>m</i> = 2: 176 (SM-PMA)	The combination of <i>m</i> :16 PMA and 16: <i>n</i> PMA forms an <i>m</i> : <i>n</i> PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
400GBASE-R 16: <i>n</i> PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	
400GAUI- <i>n</i>	<i>n</i> = 16: 120C (C2M) <i>n</i> = 8: 120E (C2M) <i>n</i> = 4: 120G (C2M) <i>n</i> = 2: 176D (C2M)	
400GBASE-R <i>n</i> :16 PMA	<i>n</i> = 4, 8, or 16: 120 (BM-PMA) <i>n</i> = 2: 176 (SM-PMA)	

176B.7 PMA instantiation for 800 Gb/s Physical Layer Implementations

The tables in this subclause summarize the forms of 800 Gb/s PMA instantiations and provide references to the clauses that define each sublayer.

176B.7.1 32:4 and 32:8 PMA instantiations for 800GBASE-R PHYs

PMA instantiations for use within a 800GBASE-R PHY are shown in Figure 176B–2, where the number of PCS lanes *q* is 32. Instantiations may be of the form 32:*n*:*p* and 32:*m*:*n*:*p*, where *p* is the number PMD lanes. Specific implementations of the PMA instantiations are defined in Table 176B–19 through Table 176B–24. These instantiations are relevant to the 800GBASE-R PHY types listed in Table 169–2 and Table 169–3, where the number of PMD lanes *p* is 4 or 8. These instantiations are also relevant to the 800GBASE-R PHY types listed in Table 169–3a.

These instantiations include C2M and C2C 800GAUI-*n* defined in Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUl lanes (*m* or *n*) is 4 or 8. Interfaces with 8 lanes require a BM-PMA and interfaces with 4 lanes require an SM-PMA. Each table specifies combinations of symbol-multiplexed interfaces (“S”) and bit-multiplexed interfaces (“B”) noted in each figure title. For example, an instantiation with a two S 800GAUI-*n* and a B PMD is denoted “SSB”.

Table 176B–19—800 Gb/s PMA 32:4:4 and 32:8:8 ($n = p$, SS or BB) instantiation

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C), 120G (C2M) $n = 4$: 176C (C2C), 176D (C2M)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	

Table 176B–20—800 Gb/s PMA 32:4:8 and 32:8:4 ($n \neq p$, SB or BS) instantiation

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C), 120G (C2M) $n = 4$: 176C (C2C), 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	The combination of n :32 PMA and 32: p PMA forms an n : p PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: p PMA	$p = 8$: 173 $p = 4$: 176	

Table 176B–21—800 Gb/s PMA 32:4:4:4 and 32:8:8:8 ($m = n$, $n = p$, SSS or BBB) instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C) $n = 4$: 176C (C2C)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	

Table 176B–22—800 Gb/s PMA 32:4:4:8 and 32:8:8:4 ($m = n$, $n \neq p$, SSB or BBS) instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C) $n = 4$: 176C (C2C)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	The combination of n :32 PMA and 32: p PMA forms an n : p PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: p PMA	$p = 8$: 173 $p = 4$: 176	

Table 176B–23—800 Gb/s PMA 32:4:8:8 and 32:8:4:4 ($m \neq n$, $n = p$, SBB or BSS) instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: m PMA	$m = 8$: 173 $m = 4$: 176	
800GAUI- m	$m = 8$: 120F (C2C) $m = 4$: 176C (C2C)	
800GBASE-R m :32 PMA	$m = 8$: 173 $m = 4$: 176	The combination of m :32 PMA and 32: n PMA forms an m : n PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	

Table 176B–24—800 Gb/s PMA 32:4:8:4 and 32:8:4:8 ($m \neq n$, $n \neq p$, SBS or BSB) instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: m PMA	$m = 8$: 173 $m = 4$: 176	
800GAUI- m	$m = 8$: 120F (C2C) $m = 4$: 176C (C2C)	
800GBASE-R m :32 PMA	$m = 8$: 173 $m = 4$: 176	The combination of m :32 PMA and 32: n PMA forms an m : n PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	The combination of n :32 PMA and 32: p PMA forms an n : p PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: p PMA	$p = 8$: 173 $p = 4$: 176	

176B.7.2 32:32 PMA instantiations for 800GMII Extender and 800GBASE-R PHYs

PMA instantiations for use within an 800GMII Extender (see Clause 171) are shown in Figure 176B–3, where the number of PCS lanes q is 32. Instantiations may be of the form 32: n :32 and 32: m : n :32. These instantiations are also relevant to the 800GBASE-R PHY type defined in Clause 185 and shown (with Inner FEC) in Figure 176B–2. Specific implementations of the PMA instantiations are defined in Table 176B–25 through Table 176B–27.

These instantiations include C2M and C2C 800GAUI- n defined in Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUI lanes (m or n) is 4 or 8. Interfaces with 8 lanes require a BM-PMA and interfaces with 4 lanes require an SM-PMA. Each table specifies combinations of symbol-multiplexed interfaces (“S”) and bit-multiplexed interfaces (“B”) noted in each figure title. For example, an instantiation with a one S 800GAUI- n and one B 800GAUI- n is denoted “SB” or “BS”.

Table 176B–25—800 Gb/s 32:4:32 and 32:8:32 PMA (S or B) instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C), 120G (C2M) $n = 4$: 176C (C2C), 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	

Table 176B–26—800 Gb/s 32:8:8:32 and 32:4:4:32 ($n = m$, BB or SS) PMA instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120F (C2C) $n = 4$: 176C (C2C)	
800GBASE-R n : n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	

Table 176B–27—800 Gb/s PMA 32:4:8:32 and 32:8:4:32 ($n \neq m$, SB or BS) instantiations

Sublayer	Clause/Annex	Notes
800GBASE-R 32: m PMA	$m = 8$: 173 $m = 4$: 176	
800GAUI- m	$m = 8$: 120F (C2C) $m = 4$: 176C (C2C)	
800GBASE-R m :32 PMA	$m = 8$: 173 $m = 4$: 176	The combination of m :32 PMA and 32: n PMA forms an m : n PMA, which converts between bit-multiplexing and symbol-multiplexing (see 176B.3).
800GBASE-R 32: n PMA	$n = 8$: 173 $n = 4$: 176	
800GAUI- n	$n = 8$: 120G (C2M) $n = 4$: 176D (C2M)	
800GBASE-R n :32 PMA	$n = 8$: 173 $n = 4$: 176	

176B.8 PMA instantiations for 1.6 Tb/s Physical Layer Implementations

The tables in this subclause summarize the forms PMA instantiations for 1.6 Tb/s Physical Layer implementations.

176B.8.1 16:8 PMA instantiations for 1.6TBASE-R PHYs

PMA instantiations for use within a 1.6TBASE-R PHY are shown in Figure 176B–2, where the number of PCS lanes q is 16. Instantiations may be of the form 16: n :8 and 16: m : n :8. Specific implementations of the

PMA instantiations are defined in Table 176B–28 and Table 176B–29. These instantiations are relevant to the 1.6TBASE-R PHY types listed in Table 174–3 and Table 174–2, where the number of PMD lanes p is 8.

These instantiations include C2M and C2C 1.6TAUI- n defined in Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUI lanes (m or n) is 8 or 16. All interfaces require an SM-PMA.

Table 176B–28—1.6 Tb/s 16:16:8 and 16:8:8 PMA instantiations

Sublayer	Clause/Annex	Notes
16: n PMA	176	
1.6TAUI- n	$n = 16$: 120F (C2C), 120G (C2M) $n = 8$: 176C (C2C), 176D (C2M)	
n :8 PMA	176	

Table 176B–29—1.6 Tb/s 16:16:16:8, 16:16:8:8, 16:8:16:8, and 16:8:8:8 PMA instantiations

Sublayer	Clause/Annex	Notes
16: m PMA	176	
1.6TAUI- m	$m = 16$: 120F (C2C) $m = 8$: 176C (C2C)	
m : n PMA	176	
1.6TAUI- n	$n = 16$: 120G (C2M) $n = 8$: 176D (C2M)	
n :8 PMA	176	

176B.8.2 16:16 PMA instantiations for 1.6TMII Extender

PMA instantiations for use within a 1.6TMII Extender (see Clause 171) are shown in Figure 176B–3, where the number of PCS lanes q is 16. Instantiations may be of the form 16: n :16 and 16: m : n :16. Specific implementations of the PMA instantiations are defined in Table 176B–30 and Table 176B–31.

These instantiations include C2M and C2C 1.6TAUI- n defined in Annex 120F, Annex 120G, Annex 176C, and Annex 176D, where the number of AUI lanes (m or n) is 8 or 16. All interfaces require an SM-PMA.

Table 176B–30—1.6 Tb/s 16:16:16 and 16:8:16 PMA instantiations

Sublayer	Clause/Annex	Notes
16: <i>n</i> PMA	176	
1.6TAUI- <i>n</i>	<i>n</i> = 16: 120F (C2C), 120G (C2M) <i>n</i> = 8: 176C (C2C), 176D (C2M)	
<i>n</i> :16 PMA	176	

Table 176B–31—1.6 Tb/s 16:16:16:16, 16:16:8:16, 16:8:16:16, and 16:8:8:16 PMA instantiations

Sublayer	Clause/Annex	Notes
16: <i>m</i> PMA	176	
1.6TAUI- <i>m</i>	<i>m</i> = 16: 120F (C2C) <i>m</i> = 8: 176C (C2C)	
<i>m</i> : <i>n</i> PMA	176	
1.6TAUI- <i>n</i>	<i>n</i> = 16: 120G (C2M) <i>n</i> = 8: 176D (C2M)	
<i>n</i> :16 PMA	176	